TRANSITGUARD: EDGE TELEMATICS FUEL ARCHITECTURE

Autonomous Digital Fuel Twin & Multi-Modal Asset Protection Framework for Commercial Logistics

Document Ref:	TG-LOG-2026-V2	**Author:**	Dr. Sunder Ramakrishnan
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1. System Framework & Paradigm Shift

To guarantee high scalability across large commercial logistics fleets, this architecture operates completely independent of external static road infrastructure. By eliminating high-cost physical components—specifically dual inline fuel flow meters and external weigh-in-motion highway platforms—the framework establishes a secure, vehicle-autonomous ecosystem. The core architecture uses an advanced algorithmic fuel consumption model paired with low-cost electronic physical defenses, all orchestrated by a central analytics coordinator.

THE PRODUCT CENTERPIECE: THE AI CONFIDENCE ENGINE

The core intelligence of TransitGuard resides within the **AI Confidence Engine**. Rather than generating binary alerts based on raw, isolated sensor triggers (which yield high false-positive rates due to road dynamics), the Confidence Engine acts as a multi-layered sensor fusion and arbitration layer.

It continuously processes independent telemetry inputs—volumetric shifts, powertrain dynamics, localized acceleration vectors, and structural acoustics—to calculate a real-time **Dynamic Fuel Security Score (0–100%)**. High-priority alarms and automated valve isolations are executed exclusively when mathematically calculated confidence thresholds are breached, ensuring institutional-grade alert reliability.

2. The Digital Fuel Twin Engine

Instead of physical return-line flow metering, the edge computer deploys a real-time, software-defined **Digital Fuel Twin** mapping the vehicle's exact engine thermodynamics. Pulling diagnostic streams directly from the internal CAN-bus network (J1939 protocol), the engine predicts consumption with a highly precise baseline accuracy targeting an average **Mean Absolute Percentage Error (MAPE) of ≤ 2.0%** under standard duty cycles. The model continuously synthesizes seven core parameters:

- **Powertrain Combustion Dynamics:** Real-time Engine Load percentage, Engine RPM matrix, engaged Gear ratio, and true Vehicle Speed.

- **Temporal & Spatial Constraints:** Total Cumulative Idle Duration and GPS-derived Elevation profiles to isolate grade-resistance load.
- **The Payload Verification Pipeline:** To feed accurate mass metrics into the consumption model, payload is derived through a triangulated hierarchy: primarily estimated via vehicle dynamics (engine torque output vs. linear acceleration rates checked against the 3-axis IMU), and cross-referenced with on-board axle load cells / air suspension pressure sensors and digital dispatch e-way bills.

3. Temperature-Compensated Density Validation

To resolve physical fluid expansion drift under severe operating temperatures, the bottom-mounted fluid analytics probe integrates a dynamic software correction model:

- **Thermal Normalization:** Raw density readings are paired with localized RTD sensor outputs and mathematically normalized to a standard 15°C reference curve via the equation: $\rho_T = \rho_0 - \beta(T - T_0)$.
- **Reference Boundaries & Confidence Limits:** Following fuel filling verification, the system captures a unique baseline chemical fingerprint. Tight statistical confidence limits (±0.5%) are maintained during transit. Any breach of this envelope indicates liquid fuel adulteration (e.g., introduction of kerosene, water, or low-grade industrial solvents).

4. Multi-Modal Security Integration Matrix

Security Layer	Sensor Input Infrastructure	Confidence Engine Context & Defense Metric
Volumetric Shift	High-Resolution Capacitive Tank Rod	Detects sustained low-volume theft events with litre-scale sensitivity after dynamic slosh compensation.
Acoustic & Vibration	Chassis-Bonded Piezo-Electric Sensors	Identifies structural frequencies of drilling, sawing, or mechanical tank breach while ignoring normal road harmonics.
Position Disruption	Magnetic Tamper Switches	Logs immediate alarms if the physical fuel tank or structural retention bands are unbolted or dropped from the chassis.
Wiring Sabotage	Active Low-Voltage Cable Loops	Maintains a continuous electrical loop; logs immediate alerts if sensor communication lines are cut or disconnected.